American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

CyberLayer: Multi-Layered Cyber Threat Defense

A Software Engineering Project Submitted By

Semester: Summer_21_22		**Section:**	**Group Number:**	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
01	Monjurul Islam Nimon	23-50004-1	30%	
02	Prihtiraj Modak	23-50007-1	25%	
03	Ariana Rahman Lazuk	23-50019-1	15%	
04	Md. Shahariyar Kabir Shanto	23-50028-1	15%	
05	Mst. Kamrun Nahr Omi	23-50483-1	15%	

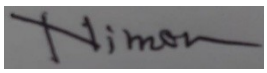
The project will be Evaluated for the following Course Outcomes

CO3: *Select* appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]	
Evidence of Argumentation regarding process model selection	[5Marks]	
Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]	
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: Monjurul Islam Nimon
Student ID: 23-50004-1
Contribution in Percentage (%): 30%
Contribution in the Project:

- Project Objective, Functional Requirements, Process Model, Impact Identification



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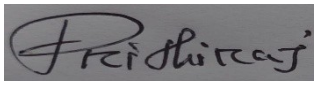
Signature of the Student

Student Name: Prithiraj Modak
Student ID: 23-50007-1
Contribution in Percentage (%): 25%
Contribution in the Project:

- Basic Functionalities, Functional Requirements, Project Role Identification



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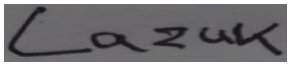
Signature of the Student

Student Name: Ariana Rahman Lazuk
Student ID: 23-50019-1
Contribution in Percentage (%): 15%
Contribution in the Project:

- Background, Reference, Non-Functional Requirements



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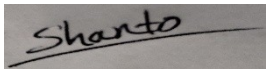
Signature of the Student

Student Name: Md. Shahariyar Kabir Shanto
Student ID: 23-50028-1
Contribution in Percentage (%): 15%
Contribution in the Project:

- Target Group, Non-Functional Requirements



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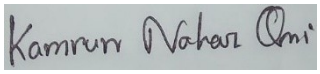
Signature of the Student

Student Name: Mst. Kamrun Nahar Omi
Student ID: 23-50483-1
Contribution in Percentage (%): 15%
Contribution in the Project:

- User Benefits, Constrain



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1.**PROJECT PROPOSAL**

1.1 Background to the Problem

As cyber threats continue to rise, businesses must adopt strong cybersecurity systems to defend against various attack types, such as DDoS, SQL Injection, and Phishing. Many existing systems either focus on general anomaly detection or specific threats but fail to address both effectively. Traditional defenses often leave gaps, leading to missed threats or excessive false positives. A multi-layered approach combining broad anomaly detection with targeted defenses is crucial. Anomaly detection identifies unusual behaviors, while specific modules target high-impact threats. CyberLayer integrates both into a single framework, offering comprehensive protection against general and specific cyber threats in real-time.

Root Cause of the Problem Importance of Considering the Problem

The increasing frequency and complexity of cyberattacks, such as DDoS, SQL Injection, and Phishing, present significant challenges to digital security. The root cause lies in the rapid evolution of threats that bypass traditional signature-based detection. High false positives and negatives reduce effectiveness, while the lack of real-time monitoring allows attackers to exploit vulnerabilities before action is taken. Additionally, human errors, especially in phishing, further complicate cybersecurity.

Addressing this problem is vital as cyberattacks compromise sensitive data, causing financial loss, identity theft, and operational disruptions. Organizations must comply with regulations like GDPR, HIPAA, and ISO 27001, and failing to do so can result in legal penalties. The rise of AI-powered attacks makes traditional defenses ineffective, highlighting the need for more advanced and adaptive security solutions. As digitalization grows, robust cybersecurity is crucial to prevent data breaches and ensure business continuity.

CyberLayer is designed to address these challenges by integrating a dual-layer security system that combines anomaly detection and targeted attack identification. This approach ensures real-time monitoring, reduces false positives and false negatives, and enhances the system's ability to detect and mitigate emerging cyber threats effectively.

1.2 **Solution to the Problem**

1. **Project Objective:** CyberLayer aims to enhance cybersecurity by providing a dual-layer security system that integrates anomaly detection with targeted attack identification. The primary objective is to develop an intelligent, adaptive security solution capable of detecting and mitigating cyber threats such as DDoS attacks, SQL injection, and phishing attempts in real-time. The system is designed to minimize false positives and false negatives, ensuring accurate threat detection and efficient mitigation strategies.

2. **Proposed Solutions:** CyberLayer's solution adopts a hybrid model that combines Adaptive Software Development (ASD) and the Spiral Model, ensuring flexibility, quick adaptation to evolving threats, and thorough validation throughout development. Key features include anomaly-based threat detection powered by machine learning to identify unusual network patterns, and a second layer of targeted attack identification for known threats like DDoS, SQL injection, and phishing. The system provides real-time alerts and mitigation strategies to minimize damage. Additionally, its scalable and adaptive framework allows seamless integration with existing security infrastructures, ensuring efficient deployment and adaptability.
3. **Appropriateness and Feasibility:** The proposed solution effectively tackles cybersecurity challenges by combining machine learning-based anomaly detection with rule-based attack identification, enhancing accuracy and reducing false alerts. CyberLayer is feasible due to its use of existing datasets, compatibility with standard infrastructures, and scalability through cloud resources. Its user-friendly interface ensures minimal training for security personnel, enabling efficient threat monitoring and response.
4. **Societal Impact:** CyberLayer enhances security and trust by protecting personal and organizational data from cyber threats, preventing identity theft, financial fraud, and business disruptions. It promotes ethical data handling and privacy rights through regulatory compliance, enabling safe online interactions and contributing to a more secure digital society.
5. **Target Users and Benefits:** CyberLayer offers key benefits to various user groups. It helps organizations and enterprises protect sensitive data and maintain business continuity. Cybersecurity analysts gain efficiency through real-time insights and automated responses. Government institutions secure critical infrastructure and citizen data, while financial institutions prevent fraud and phishing. Overall, users benefit from reduced cybersecurity risks, lower operational costs, and better compliance with legal regulations.
6. **Scientific Contribution:** CyberLayer advances cybersecurity by combining anomaly detection with rule-based identification, improving threat detection accuracy. Its machine learning approach reduces false positives, enhancing security system efficiency. Additionally, CyberLayer offers a scalable solution that integrates seamlessly with existing infrastructures, providing robust protection against evolving cyber threats.
7. **Literature Review:** Existing cybersecurity studies mainly focus on anomaly detection or rule-based identification separately. CyberLayer improves on this by combining both methods for more accurate threat detection and fewer false alerts. Influential studies include Smith et al. (2021) on machine learning in cybersecurity, Johnson & Wang (2020) on real-time threat detection, and Lee et al. (2019) on adaptive security frameworks, which helped shape CyberLayer's hybrid approach.
8. **Comparison with Existing Solutions:** CyberLayer improves upon existing cybersecurity solutions such as Snort, a signature-based intrusion detection system, Suricata, a high-performance threat detection engine, and Darktrace, which uses AI-driven anomaly detection. By combining real-time anomaly detection with targeted attack identification,

CyberLayer offers a more adaptive and scalable solution, with reduced false positives and negatives. Additionally, it seamlessly integrates with modern security frameworks, enhancing overall effectiveness in protecting against evolving cyber threats.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model : Hybrid model(ADS+Spiral)

Analysis of the Nature and Environment of CyberLayer

The **CyberLayer** software is a cybersecurity system designed to detect and mitigate a wide range of cyber threats, including DDoS attacks, SQL injection, and phishing. The software aims to provide real-time protection and monitoring, ensuring that security threats are promptly identified and neutralized. Given the critical nature of cybersecurity applications, the system must not only be adaptable to the ever-evolving landscape of cyber threats but also be capable of performing rigorous security validations at every stage of development.

Environment of Software

Cybersecurity systems like **CyberLayer** operate in a dynamic environment where cyber threats evolve rapidly, requiring constant updates and adaptations to ensure that the software remains effective against new attack vectors. Additionally, the system must meet stringent security and performance requirements, such as real-time monitoring, high availability, and minimal downtime. The development environment also involves collaboration between security analysts, software developers, and system administrators, with a strong focus on rapid prototyping, testing, and continuous feedback.

Arguments for Selecting Hybrid Model (ASD + Spiral)

Based on the nature and environment of the software, the most suitable development methodology for CyberLayer is a Hybrid Model that combines Adaptive Software Development (ASD) and the Spiral Model. This Hybrid Model offers the flexibility, adaptability, and rigorous validation required for a cybersecurity-focused system. It's demonstrate below:

1. **Adaptability to Evolving Threats (ASD)**: Adaptive Software Development (ASD) is ideal for cybersecurity systems due to its flexibility and iterative approach. It allows quick adaptation to new threats and rapid deployment of updates, ensuring the system can evolve and stay ahead of emerging risks. ASD also enables prompt response to incidents and continuous improvement based on user feedback.

2. **Comprehensive Risk Management and Security Validation (Spiral Model)**: The Spiral Model is ideal for cybersecurity systems like CyberLayer due to its focus on risk management and iterative validation. It allows for continuous risk assessment, thorough testing, and early identification of vulnerabilities, ensuring the system remains secure and resilient against attacks. This approach is crucial for meeting high-security standards.
3. **Combining Flexibility with Structured Security**: The Hybrid Model blends ASD's adaptability with the Spiral Model's risk management, ensuring CyberLayer is both flexible and secure. This approach allows quick adaptation to new threats while integrating security at each stage. Unlike Waterfall, which lacks flexibility, or Agile, which may compromise risk management, the Hybrid Model provides a balanced solution for cybersecurity.

Evidence of Success in Cybersecurity Projects:

- Adaptive Software Development (ASD) emphasizes iterative cycles and continuous adaptation, making it well-suited for environments where requirements evolve rapidly, such as cybersecurity.[1]
- The Spiral Model integrates risk analysis at each phase, aligning with the need for rigorous security validation in cybersecurity projects.[2]
- Research indicates that engaging security measures within each iteration of the Spiral Model enhances the system's resilience against emerging threats. [3]
- The Spiral Model's focus on risk management has proven beneficial in developing secure systems, as it allows for continuous assessment and mitigation of security risks throughout the development process. [4]

Reference:

1. Adaptability to Evolving Threats (ASD):ThinkPalm Technologies, "All About Adaptive Software Development (ASD)," *ThinkPalm*, 2021. [Online]. Available: https://thinkpalm.com/blogs/all-about-adaptive-software-development-asd
2. Comprehensive Risk Management and Security Validation (Spiral Model): GeeksforGeeks, "Software Engineering - Spiral Model," *GeeksforGeeks*, 2023. [Online]. Available:https://www.geeksforgeeks.org/software-engineering-spiral-model/Combining.
3. Flexibility with Structured Security: ScienceDirect, "A Study on Spiral Model in Software Engineering," *ScienceDirect*, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0268401213000418
4. Success in Cybersecurity with the Spiral Model: Groove Technology, "Spiral Model in Software Development," *Groove Technology*, 2021. [Online]. Available: https://groovetechnology.com/blog/software-development/spiral-model/

2.2 Project Role Identification and Responsibilities

1. Project Manager: Oversees the overall project, facilitates communication between team members, manages the timeline, and ensures the project stays on track and within scope. Ensures proper coordination between different roles and stakeholders.

2. Frontend Developer: Designs and implements the user interface of the *CyberLayer* system, ensuring it is intuitive and user-friendly. Uses modern frameworks such as React or Angular for responsive web design.

3. Backend Developer: Develops the core backend services of the *CyberLayer* system, implementing essential functionality like attack detection algorithms, security modules, and database integration. Uses programming languages like Python or Java for server-side logic.

4. Security Architect: Designs the system's overall security framework and architecture. Ensures that security best practices are applied throughout the development process. They focus on setting up secure communication protocols, secure data storage, and creating threat models.

5. Cybersecurity Specialist: Focuses on developing and implementing cybersecurity features for the system, such as anomaly detection, real-time monitoring, and integration of the latest security technologies. Ensures the software is protected against potential vulnerabilities.

6. Compliance Officer: Ensures that the system adheres to industry standards, legal regulations (such as GDPR, HIPAA, or CCPA), and cybersecurity frameworks (e.g., NIST). Works closely with the project manager and security team to ensure compliance in all phases of development.

7. Data Scientist (for AI/ML Models): Works with machine learning algorithms for attack detection, anomaly detection, and other AI-driven features in the system. Ensure that the models used are accurate, efficient, and updated regularly.

8. Database Engineer: Manages and optimizes the database system that stores user data, attack logs, and system information. Ensures high performance, security, and scalability of the database, likely using SQL or NoSQL technologies.

9. Quality Assurance (QA) Tester: Conducts extensive testing of the system to identify bugs, vulnerabilities, and performance issues. Ensures all security features, such as anomaly detection and alerting, function as intended.

10. UX/UI Designer: Designs the interface and user experience of the *CyberLayer* system, ensuring ease of use, accessibility, and user engagement. Focuses on creating intuitive dashboards for security alerts and reports.

Impact Identification:

1. Security Impact: CyberLayer's dual-layer approach improves threat detection by combining anomaly detection with targeted attack identification, reducing false positives. It minimizes attack surfaces and enables proactive security through real-time monitoring, preventing threats before escalation.

2.Business Impact: CyberLayer helps reduce security incidents, cutting operational costs linked to data breaches and downtime. It also protects a company's reputation by preventing cyberattacks, building trust with customers, and ensuring business continuity.

3.**User Impact:** Users benefit from better privacy and security, especially against phishing and data theft. The system's user-friendly desktop interface ensures easy adaptation, while its seamless background operation minimizes disruptions to performance.

4.Technical Impact: CyberLayer ensures system efficiency by processing data quickly without bottlenecks. Its scalable design meets the security needs of organizations of all sizes, ensuring robust protection without compromising performance.

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	**Satisfactory (3)**	**Good (4)**	**Excellent (5)**	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	Some errors in spelling and grammar. Some problems of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	Few errors in spelling and grammar. Presents most of the details in a logical flow of organization in definition, details, and example.	Project report is complete and No errors in spelling and grammar. Consistently presents a logical and effective organization of definition, details, and real-life example of the topic.	
				Acquired marks:	
				CO Pass / Fail:	